PRELIMINARY MEDICAL REPORT - LESION ANALYSIS
AI-CDSS - Artificial Intelligence Clinical Decision Support System

Synthetic example generated for the public code repository accompanying CBEB 2026 submission #35, "An AI-Enabled Digital Healthcare Workflow: Integrating Deep Learning and Large Language Models for Interpretable Wound Triage." Patient identity, history, and clinical text below are fictitious.

Issue date: 08/28/2026 at 20:25

1. PATIENT INFORMATION

Name: Jose da Silva (synthetic)
Age: 58 years old
Sex: M
Diabetes Type: Type 2
Document: Not informed
Medical History: Type 2 diabetes mellitus, peripheral neuropathy, prior lower-limb surgical intervention.
Medications: Metformin
Allergies: None reported

2. LESION ANALYSIS

2.1 Lesion 1:

Classification: D - Diabetic Ulcer (36.74%)
File: lesion_plantar_foot.jpg
Probability Distribution:

Classification	Probability	Description
D	36.74%	Diabetic Ulcer
V	24.10%	Venous Ulcer
S	18.42%	Surgical Wound
P	12.05%	Pressure Ulcer

BG	5.11%	Background
N	3.58%	Normal Skin

Formal Analysis:

The plantar surface of the foot shows a well-circumscribed lesion consistent with a diabetic ulcer, with peri-lesional erythema and mild tissue maceration. The classifier assigns 36.74% confidence to this class, indicating substantial but not conclusive certainty; the venous and surgical-wound classes carry non-negligible probability mass (24.10% and 18.42%, respectively) and should be considered in the differential diagnosis. Given the patient's history of type 2 diabetes and peripheral neuropathy, a diabetic ulcer is clinically plausible. Confirmatory clinical examination, including vascular assessment, is recommended before treatment planning.

2.2 Lesion 2:

Classification: S - Surgical Wound (49.84%)
File: lesion_dorsal_scar.jpg
Probability Distribution:

Classification	Probability	Description
S	49.84%	Surgical Wound
D	19.33%	Diabetic Ulcer
V	14.02%	Venous Ulcer
N	9.21%	Normal Skin
P	4.87%	Pressure Ulcer
BG	2.73%	Background

Formal Analysis:

The dorsal region presents a linear, healing incision with visible suture marks, consistent with a surgical wound. The classifier assigns 49.84% confidence to this class, the highest confidence among the two lesions analyzed. Diabetic and venous ulcer classes retain moderate probability (19.33% and 14.02%), which is expected given surface texture and pigmentation overlap between healing surgical scars and chronic ulcers in this model. The lesion appears to be in an advanced healing stage, with no visible signs of dehiscence or active infection.

3. SYSTEM INFORMATION

Performance of the Classification Model:

Glossary of Metrics:
• **Precision**: Accuracy in positive predictions (precision)
• **Recall**: Ability to find all positive cases (sensitivity)
• **F1-Score**: Harmonic mean between Precision and Recall
• **Values**: 0.000 (worst) to 1.000 (best)

Class	Description	Precision	Recall	F1-Score
BG	Background	0.960	0.960	0.960
D	Diabetic Ulcer	0.850	0.739	0.791
N	Normal Skin	0.833	1.000	0.909
P	Pressure Ulcer	0.615	0.235	0.340
S	Surgical Wound	0.600	0.643	0.621
V	Venous Ulcer	0.704	0.919	0.797

Classification System: VGG16 fine-tuned convolutional neural network
Purpose: Auxiliary clinical decision support for initial lesion screening
Training: Multi-modal wound classification dataset (AZH Wound and Vascular Center)
Processing: Automatic visual feature analysis

4. OBSERVATIONS AND RECOMMENDATIONS

• This document constitutes a **PRELIMINARY REPORT** generated by an Artificial Intelligence system;
• **Does not replace** the evaluation of a qualified healthcare professional;
• For pressure ulcers (Class P), the system has **limited sensitivity** (recall: 23.53%).

Document generated automatically - AI-CDSS Wound Analysis System
Patient: Jose da Silva (synthetic) | ID: demo | Generated on: 08/28/2026 20:25

Example asset for the public repository accompanying CBEB 2026 submission #35: Rocha Junior, E. F.; Sanchez-Gendriz, I.; Guedes, L. A.